

Overview

- Motivation for dimensionality reduction
- Feature selection
 - Wrappers
 - Filters
 - Embedded methods
- Feature transformation
 - Principal Component Analysis (PCA)
 - Autoencoders
 - Clustering

Motivation for dimensionality reduction

Examples of large feature spaces

Predicting recurrence of lung cancer



Only a few genes actually matter!

Need small, interpretable subset to help doctors!

Motivation for dimensionality reduction

Examples of large feature spaces

Text classification

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<TITLE>AMERICAN PORK CONGRESS KICKS OFF TOMORROW</TITLE>
<DATELINE> CHICAGO, March 2 - </DATELINE><BODY>The American Pork
Congress kicks off tomorrow, March 3, in Indianapolis with 160
of the nation's pork producers from 44 member states determining
industry positions on a number of issues, according to the
National Pork Producers Council, NPPC.
Delegates to the three day Congress will be considering 26
resolutions concerning various issues, including the future
direction of farm policy and the tax law as it applies to the
agriculture sector. The delegates will also debate whether to
endorse concepts of a national PRV (pseudorabies virus) control
and eradication program, the NPPC said. A large
trade show, in conjunction with the congress, will feature
the latest in technology in all areas of the industry, the NPPC
added. Reuter
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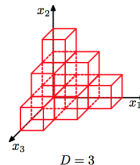
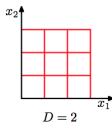
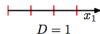
“Bag-of-Words” representation:

$x = \{0, 3, 0, 0, 1, \dots, 2, 3, 0, 0, 0, 1\}$ one entry per word!

Easily 50,000 words! Very sparse - easy to overfit!

What is curse of dimensionality

Number of cells grows exponentially as dimensionality increases



of cells r^D

r : number of divisions in each dimension

- Large number of cells, even if D is moderately large
- So to cover the whole space reasonably well, you need exponentially number of training data points

Dimensionality Reduction

Broader question

- How can we detect **low dimensional structure** in **high dimensional** data?

Motivations

- Exploratory data analysis & visualization: you can plot data now
- Compact representation: small memory/computational footprint, lossy data compression
- Robust statistical modeling: curse of dimensionality

Dimensionality Reduction

General rules of dimensionality reduction

- **Relevant** features: the features that we need to perform well
- **Irrelevant** features: the features that are unnecessary
- **Redundant** features: the features that become irrelevant in the presence of others